



VISUAL ANALYTICS: WORKING WITH BIG DATA

AB MOSCA (THEY/THEM)



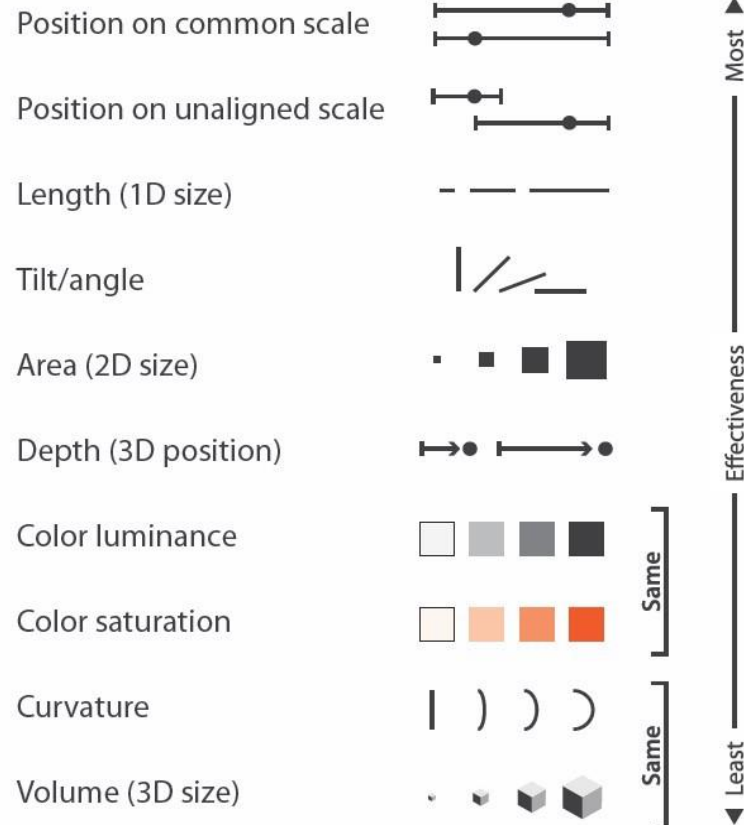
TODAY'S PLAN

- Motivation
- Faceting
- Reduction
- Focus + Context
- Big Data Storage & Access

MOTIVATION

Channels: Expressiveness Types and Effectiveness Ranks

➔ Magnitude Channels: Ordered Attributes



➔ Identity Channels: Categorical Attributes



What should we do if our data contains far more variables than visual channels?

What should we do if a variable has thousands of possible values?

FACETING

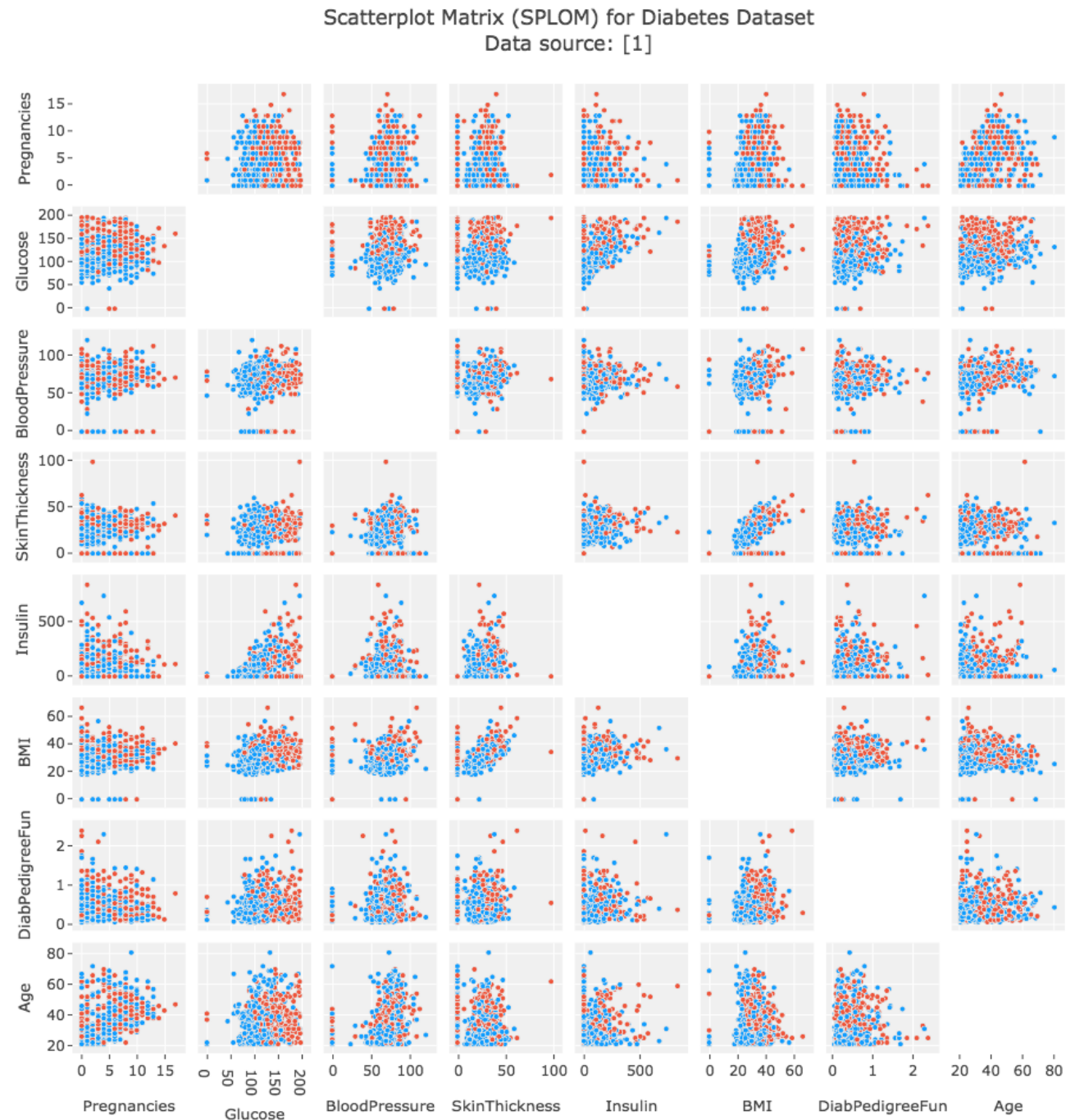
- **Faceting** refers to splitting the data and visualization into multiple views that are either juxtaposed side-by-side or layered

What examples of faceting have we already encountered?

FACETING

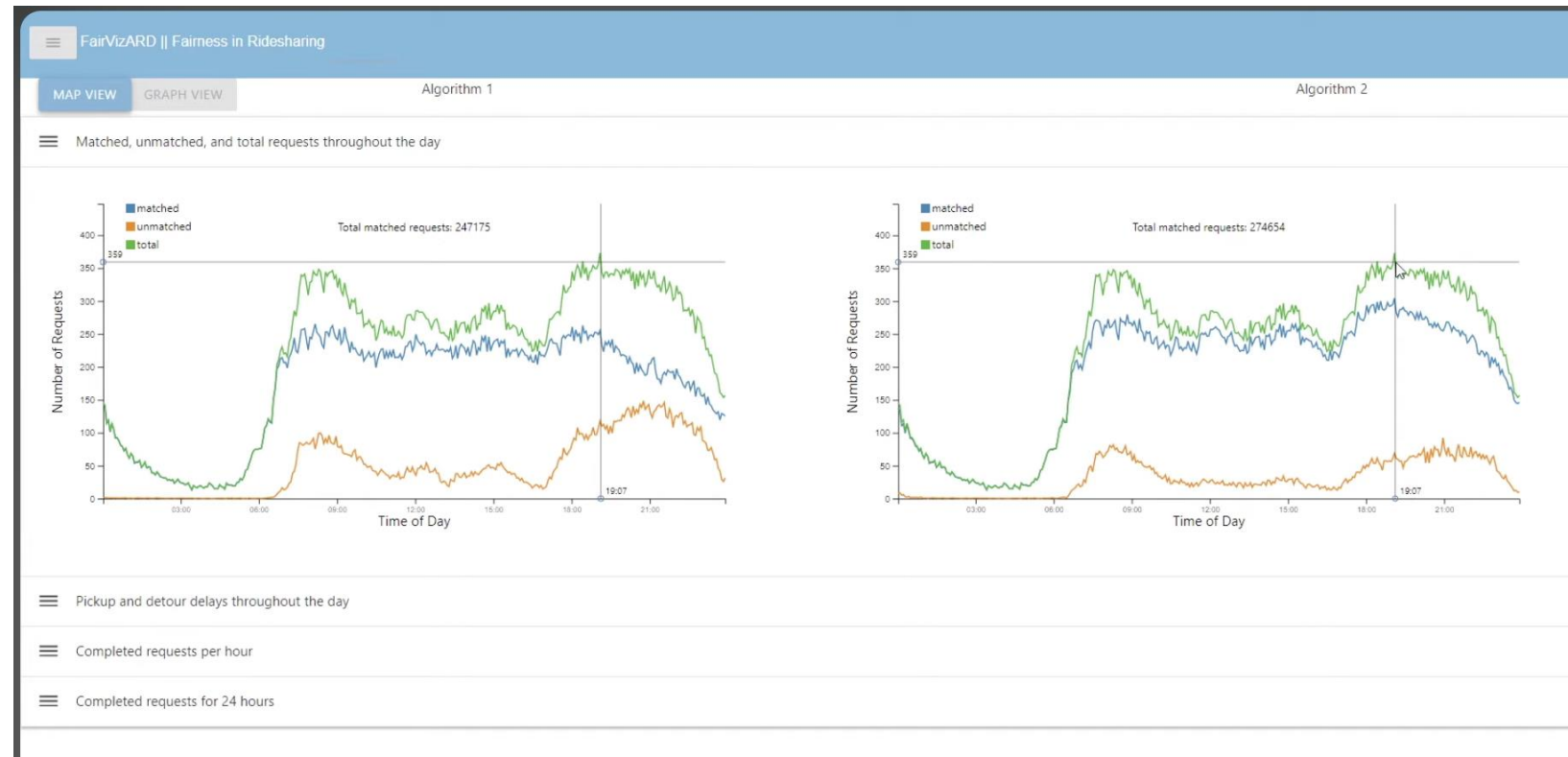
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- Ex. SPLOM – Juxtaposed same encoding

<https://medium.com/plotly/what-is-a-splom-chart-make-scatterplot-matrices-in-python-8dc4998921c3>



FACETING

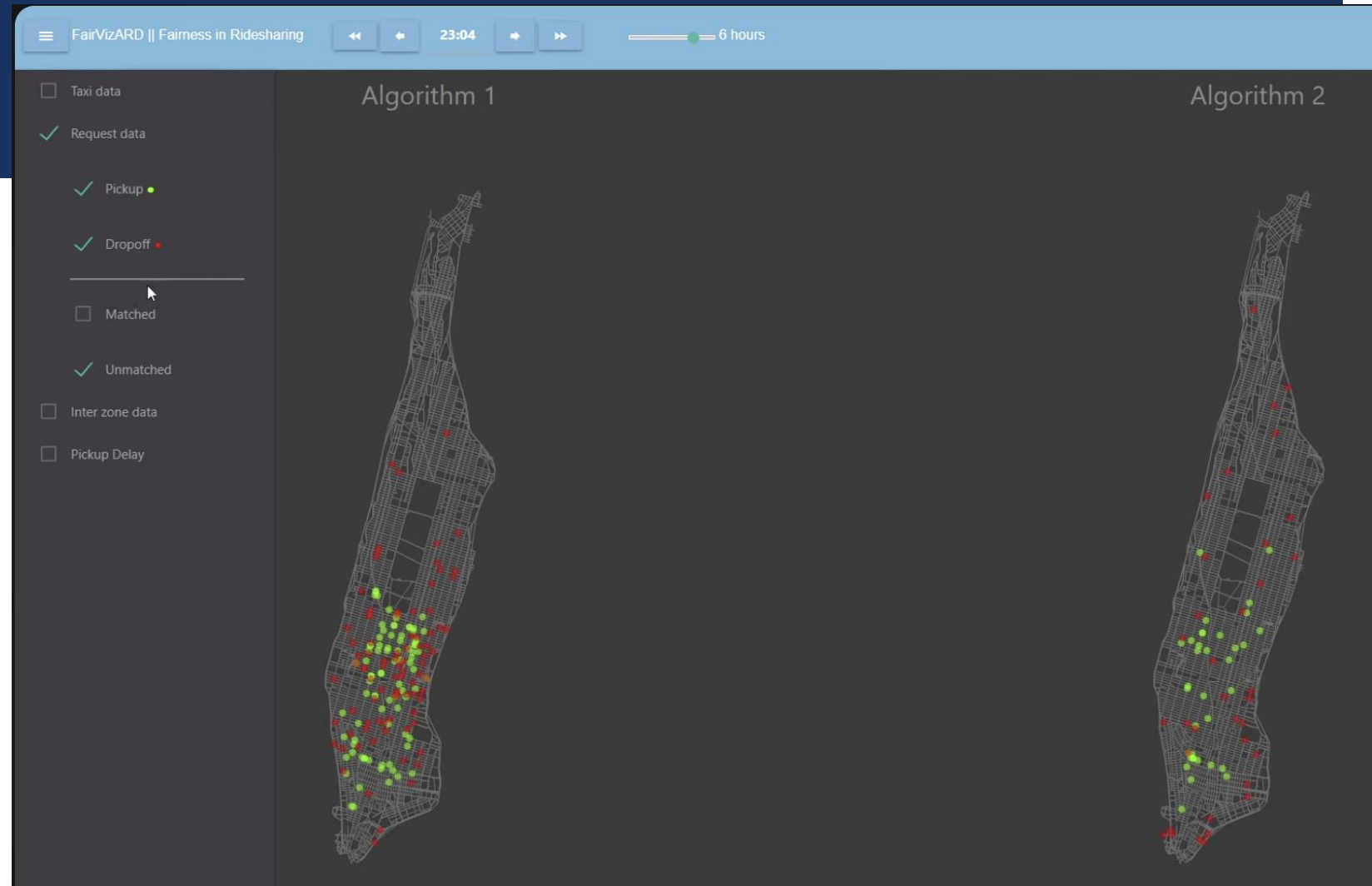
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- Ex. Juxtaposed and coordinated interaction



Kumar, A., Shah, S., Lowalekar, M., Varakantham, P., Ottley, A., & Yeoh, W. (2025). FairVizARD: A Visualization System for Assessing Multi-Party Fairness of Ride-Sharing Matching Algorithms. *arXiv preprint arXiv:2508.11770*.

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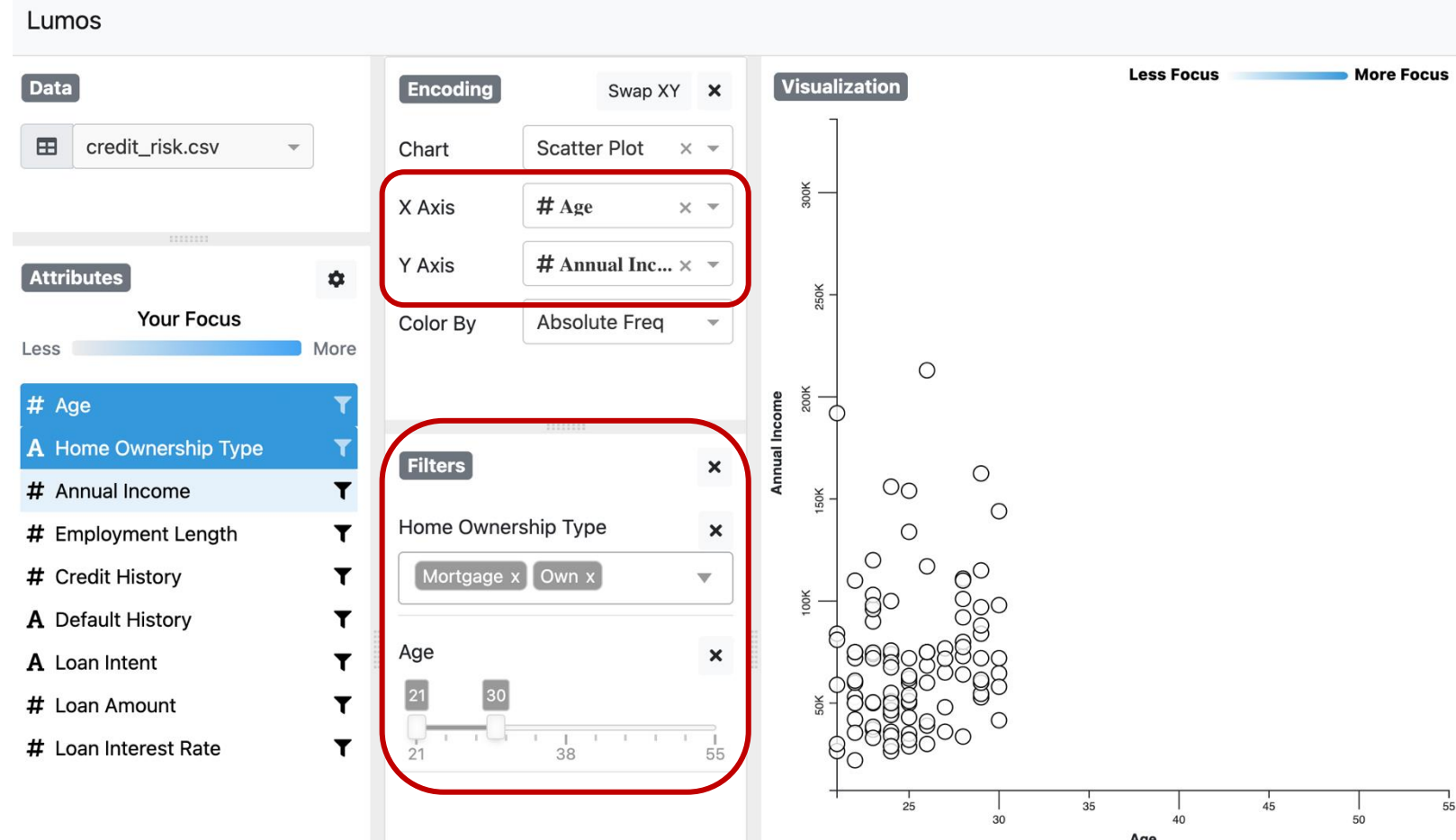
REDUCTION

- The number of variables and/or observations in a dataset can be reduced via filtering and/or aggregation

What examples of reduction have we already encountered?

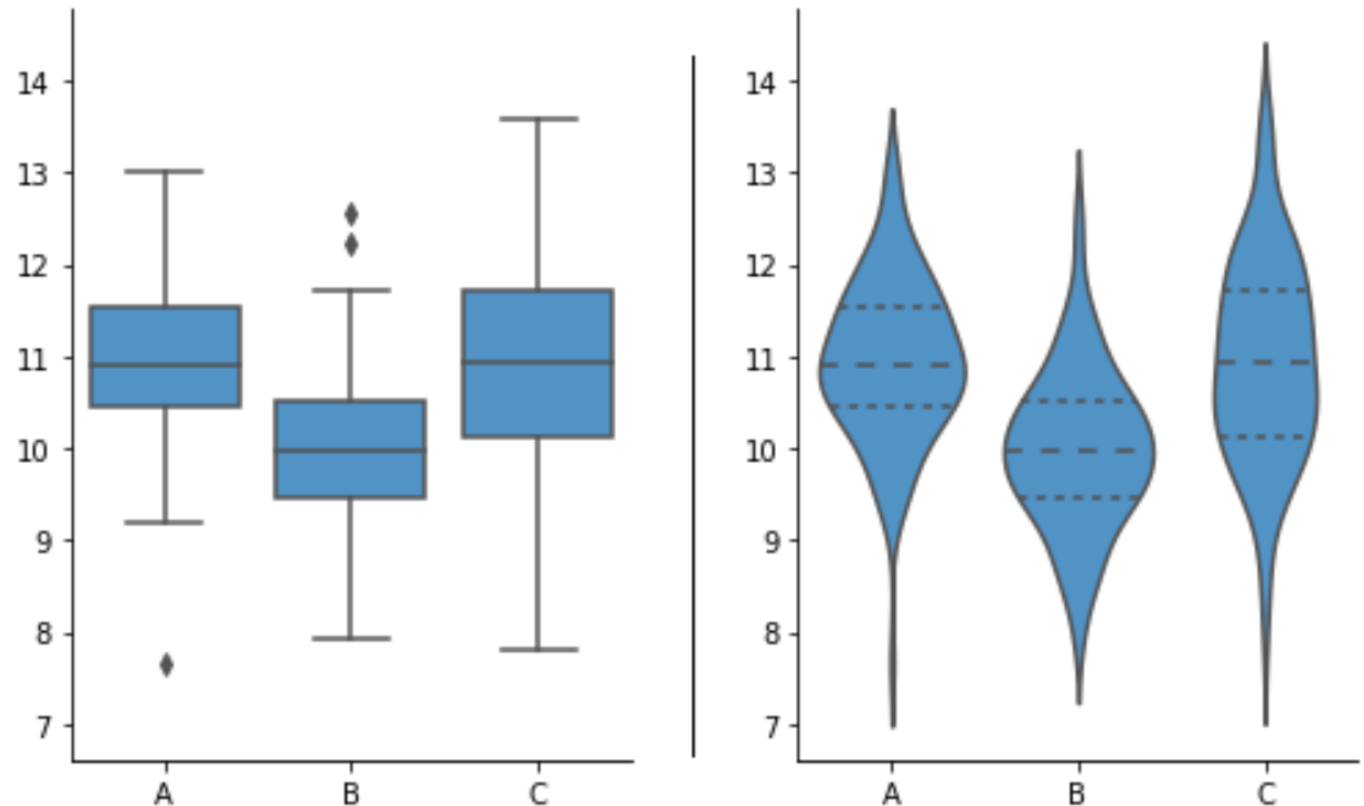
REDUCTION

- The number of variables and/or observations in a dataset can be **reduced** via filtering and/or aggregation
- Ex. Filtering variables and observations



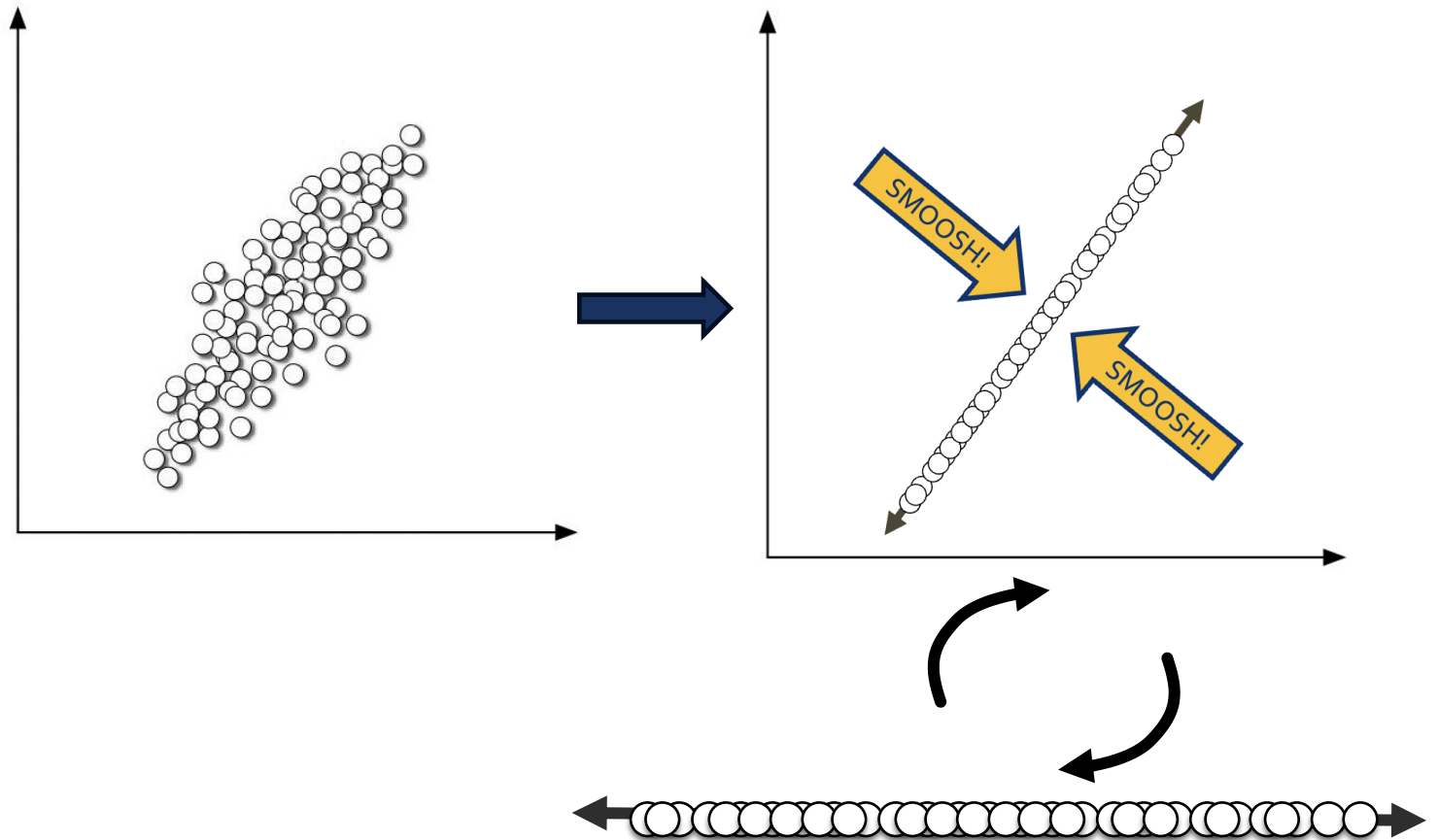
REDUCTION

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- Ex. Aggregating observations



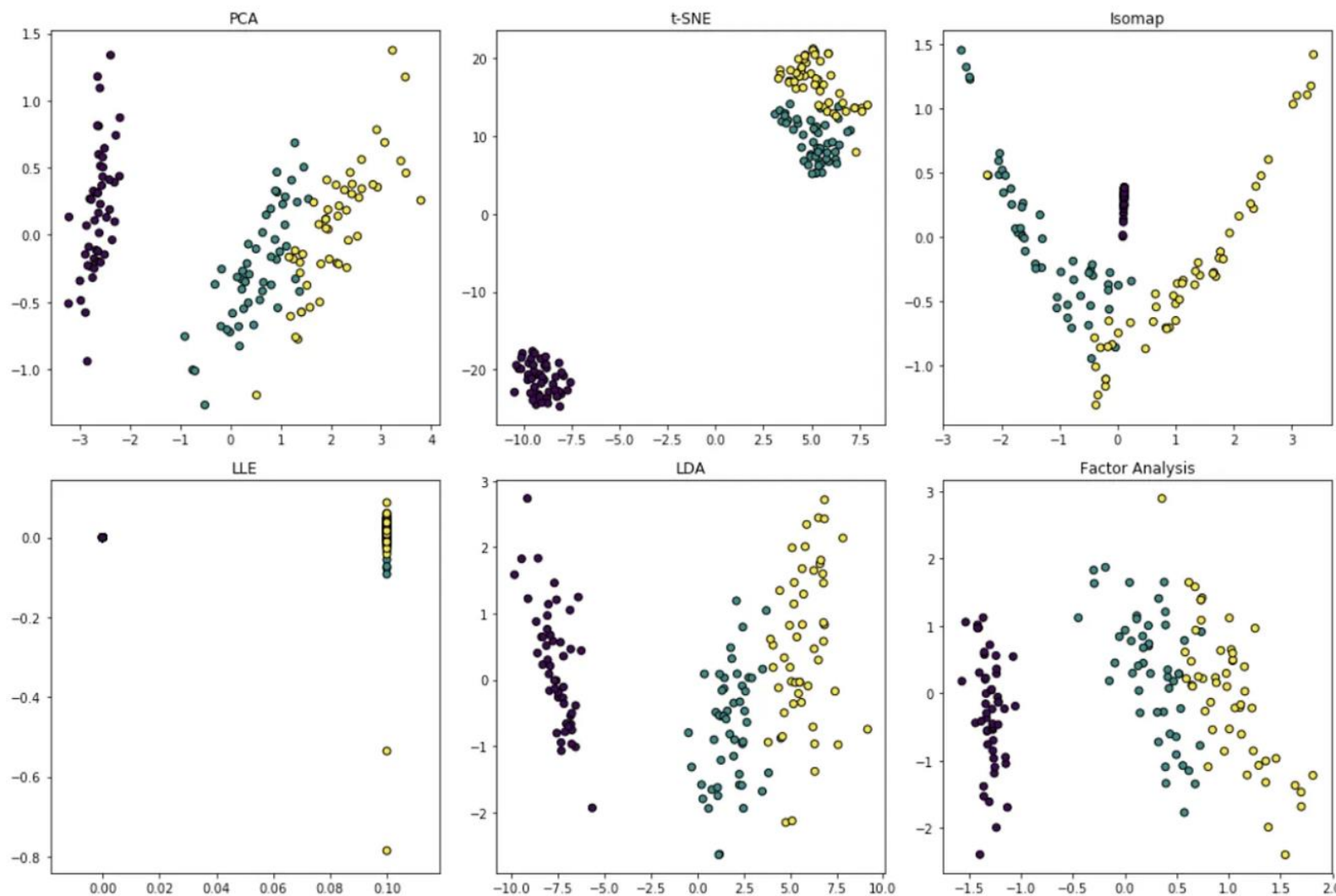
REDUCTION

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- Ex. Aggregating variables – Dimension Reduction



REDUCTION

- The number of variables and/or observations in a dataset can be reduced via filtering and/or aggregation
- Ex. Aggregating variables
 - Dimension Reduction
- Many DR techniques to choose from

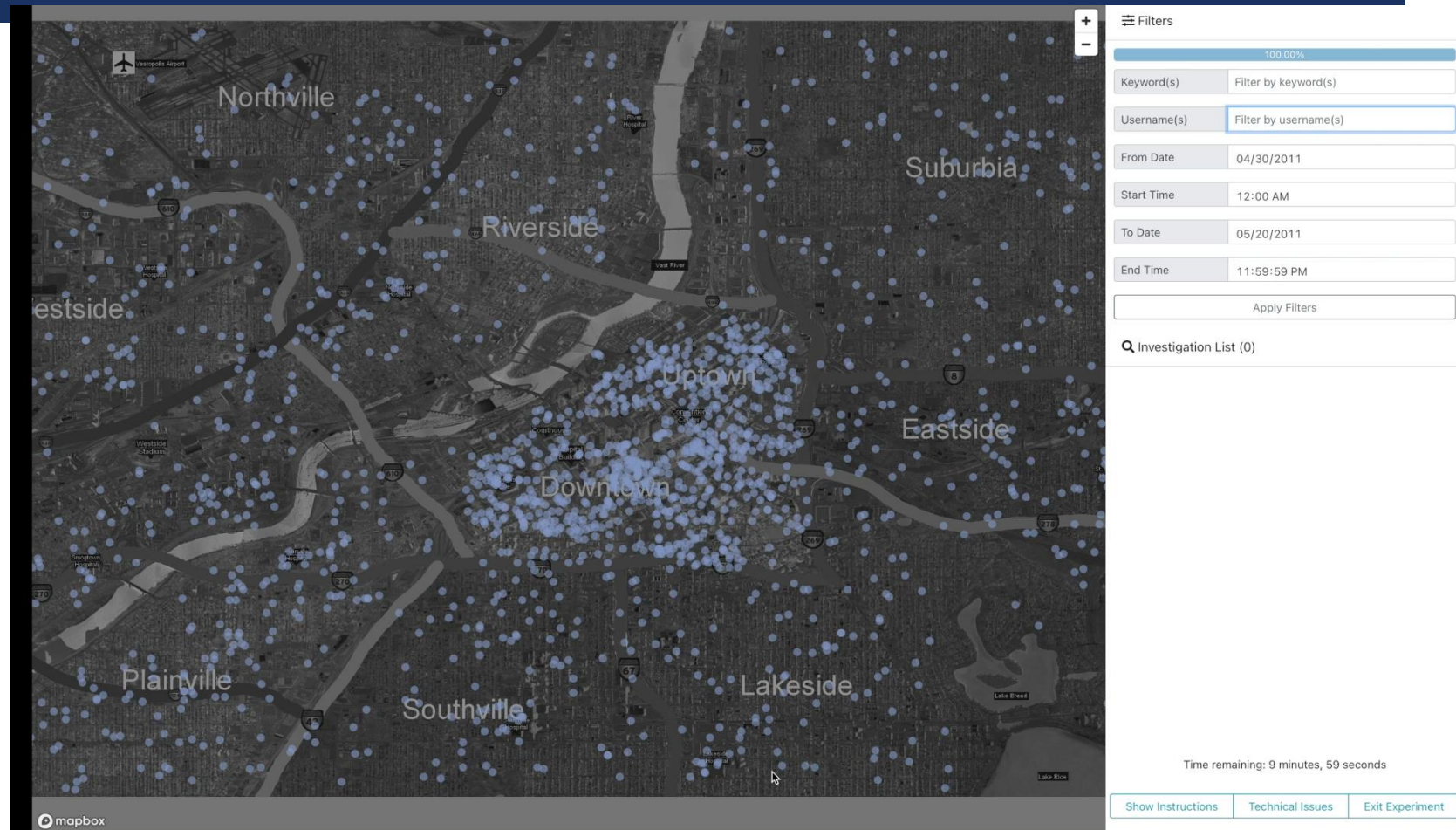


FOCUS + CONTEXT

- Focus + context refers to design options that embed detailed information about a selected subset of data within an overview that contains overall information about the dataset

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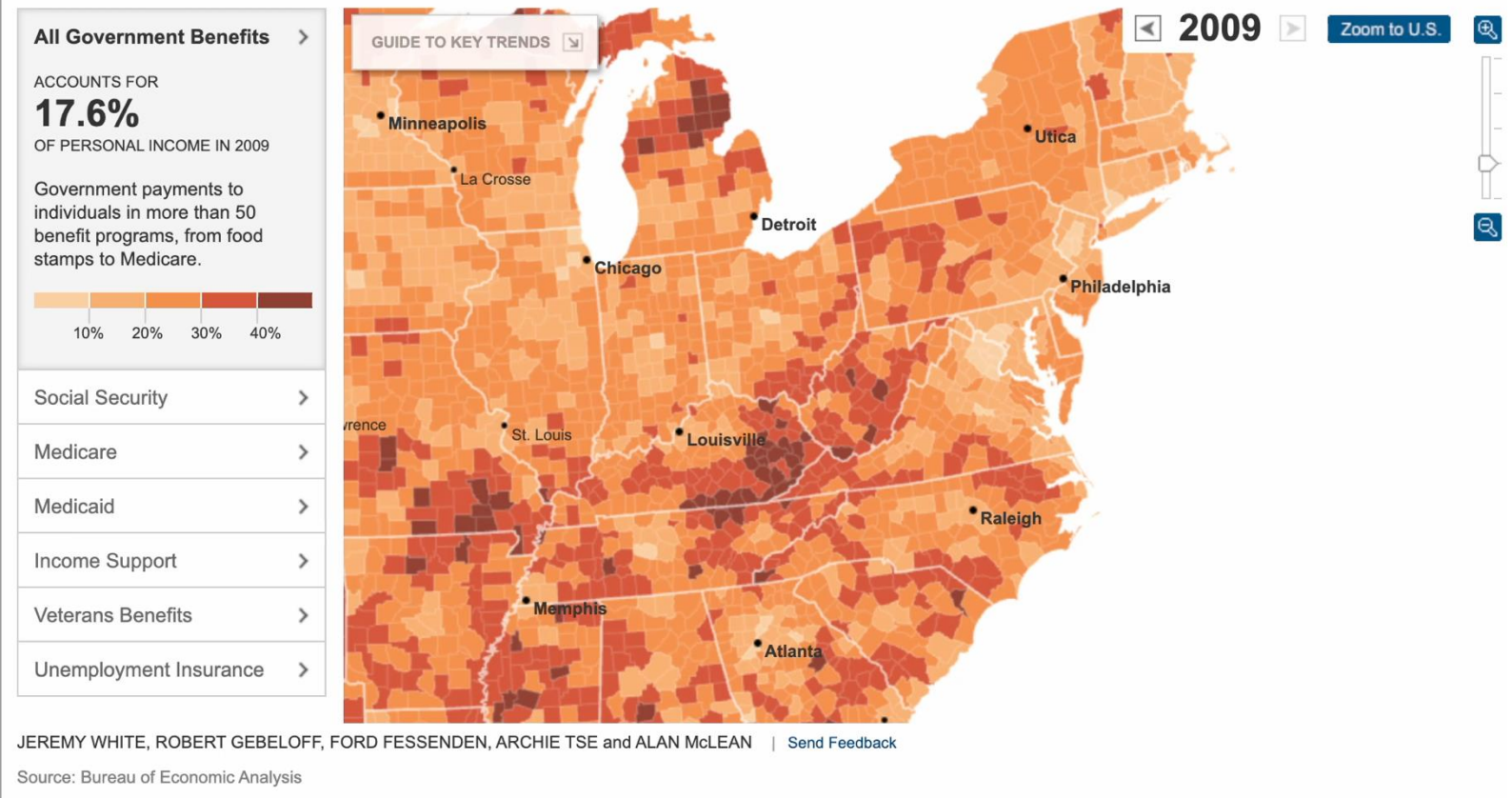
Ha, S., Monadjemi, S. and Ottley, A. (2024), Guided By AI: Navigating Trust, Bias, and Data Exploration in AI-Guided Visual Analytics. Computer Graphics Forum, 43: e15108. <https://doi.org/10.1111/cgf.15108>

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The Geography of Government Benefits

The share of Americans' income that comes from government benefit programs, like Medicare, Medicaid and Social Security, more than doubled over the last four decades, rising from 8 percent in 1969 to 18 percent in 2009. [Related Article »](#)



WHEN TO USE WHAT

- Deciding between faceting, reduction, and focus + context is a non-negligible task
- The correct choice depends on the intended user and anticipated problem space
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- Ex. What approaches does this system utilize? <https://www.mass.gov/info-details/immunizations-for-respiratory-diseases>

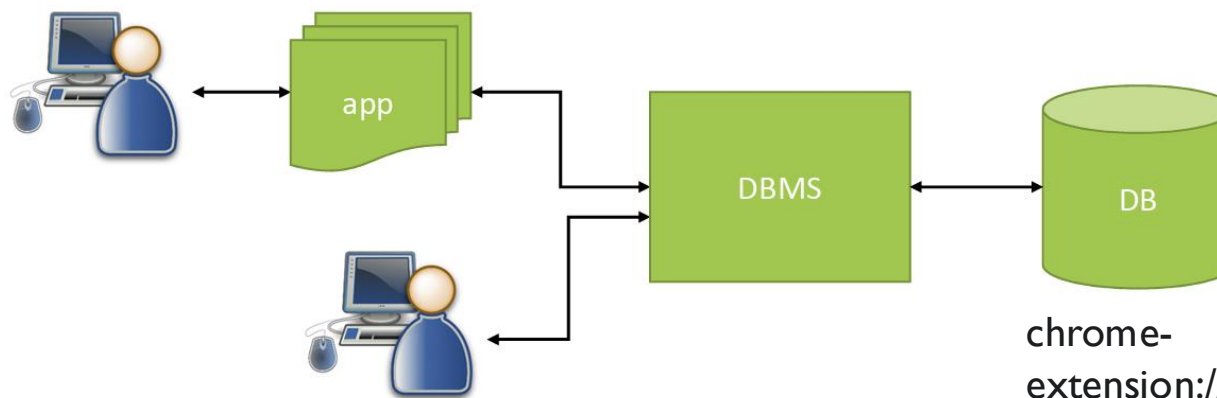
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- The correct choice depends on the intended user and anticipated problem space
- Often, we will combine multiple approaches in one visual analytic system
- Ex. What approaches does this system utilize? <https://www.mass.gov/info-details/immunizations-for-respiratory-diseases>
 - Identify two specific ways that you could re-design this tool to incorporate faceting, reduction, and/or focus + context

BIG DATA STORAGE & ACCESS

Database concepts:

- Database = Data + Base, the actual storage of information
- Database Management System (DBMS) = The software that helps manage data storage, access, and maintenance
- Database Application = An application built upon the data stored in databases

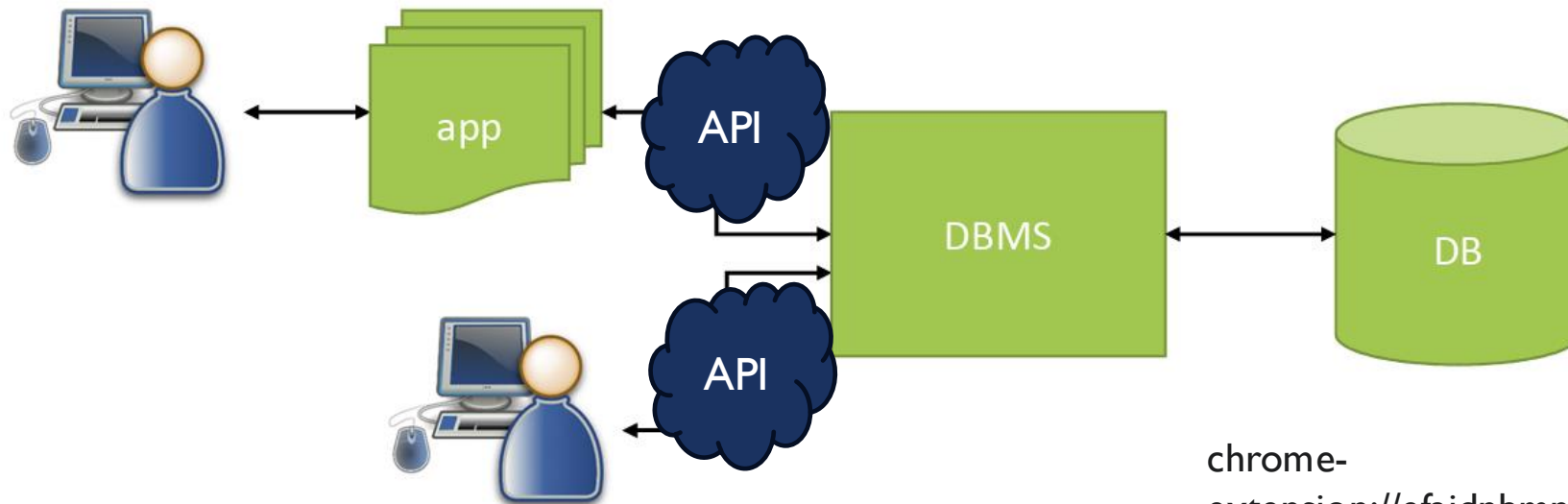


chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://www.bu.edu/tech/files/2018/06/intro2DB20180611.pdf

BIG DATA STORAGE & ACCESS

Database concepts:

- Apps and users interact with DBMSs through APIs (Application Programming Interfaces)



chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://www.bu.edu/tech/files/2018/06/intro2DB20180611.pdf

BIG DATA STORAGE & ACCESS

Database concepts:

- In this class we will largely work with relational databases

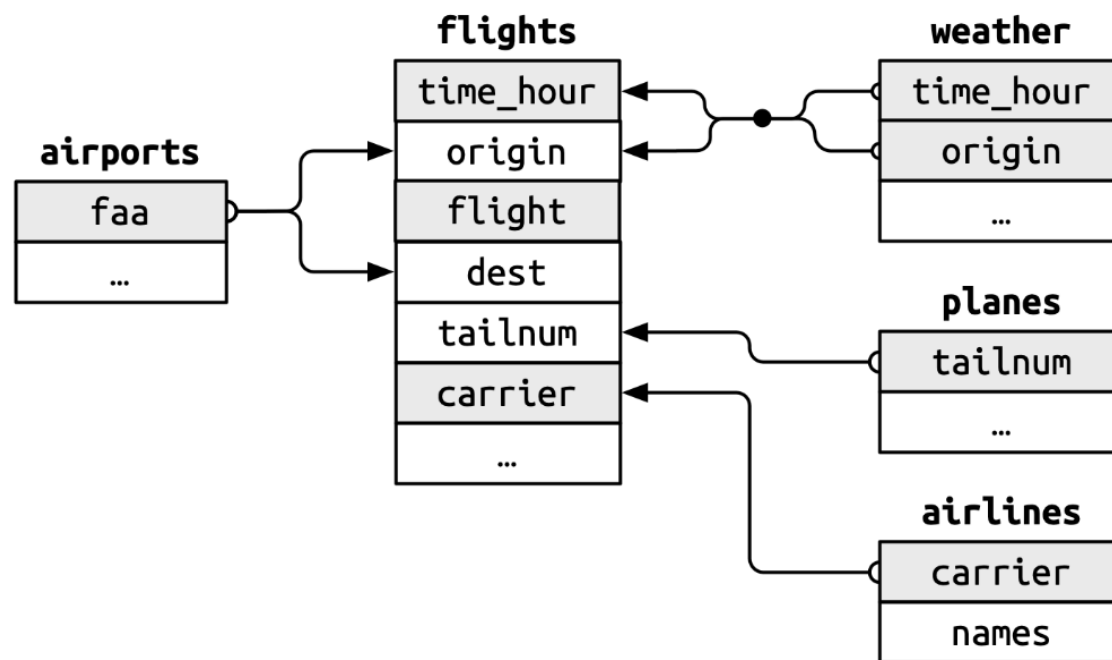


Figure 19.1: Connections between all five data frames in the nycflights13 package. Variables making up a primary key are colored grey, and are connected to their corresponding foreign keys with arrows.

BIG DATA STORAGE & ACCESS

SQL (Structured Query Language) is a standard language for accessing and manipulating databases

■ SQL statements consist of keywords. Ex.

- **SELECT** - extracts data from a database
- **UPDATE** - updates data in a database
- **DELETE** - deletes data from a database
- **INSERT INTO** - inserts new data into a database
- **CREATE DATABASE** - creates a new database
- **ALTER DATABASE** - modifies a database
- **CREATE TABLE** - creates a new table
- **ALTER TABLE** - modifies a table
- **DROP TABLE** - deletes a table
- **CREATE INDEX** - creates an index (search key)
- **DROP INDEX** - deletes an index

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SQL Syntax Examples

```
SELECT column1, column2, ...  
FROM table_name;
```

```
SELECT * FROM Customers;
```

```
SELECT DISTINCT column1, column2, ...  
FROM table_name;
```

```
SELECT Country FROM Customers;
```


BIG DATA STORAGE & ACCESS

SQL aggregate function examples:

- `MIN()` - returns the smallest value within the selected column
- `MAX()` - returns the largest value within the selected column
- `COUNT()` - returns the number of rows in a set
- `SUM()` - returns the total sum of a numerical column
- `AVG()` - returns the average value of a numerical column

Aggregate functions ignore null values (except for `COUNT(*)`).

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SQL Syntax Examples

```
SELECT MIN(column_name)
FROM table_name
WHERE condition;
```

```
SELECT MAX(column_name)
FROM table_name
WHERE condition;
```

```
SELECT MIN(Price) AS SmallestPrice
FROM Products;
```

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SQL Joins

- **(INNER) JOIN** : Returns records that have matching values in both tables
- **LEFT (OUTER) JOIN** : Returns all records from the left table, and the matched records from the right table
- **RIGHT (OUTER) JOIN** : Returns all records from the right table, and the matched records from the left table
- **FULL (OUTER) JOIN** : Returns all records when there is a match in either left or right table



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Practice Working with SQL

- Team up with someone else
- Use this interface to practice writing and running SQL queries
 - <https://www.sql-practice.com/>